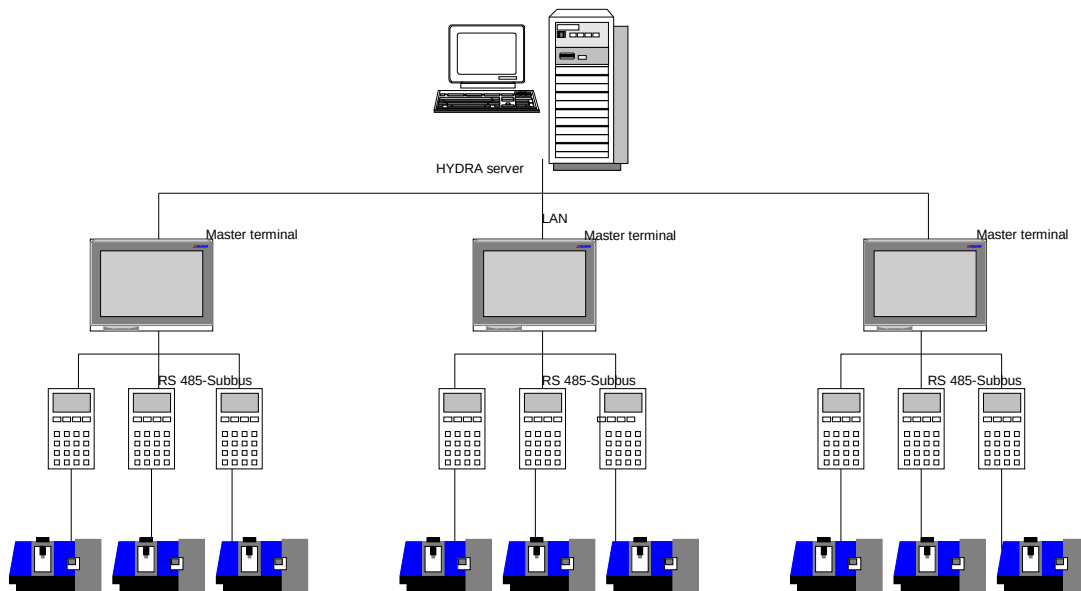


1 Master Terminal using DS-100

This document is based on the manuals dealing with the HYDRA standard Windows terminals (AIP). Consequently, this document only describes the special features if used as master terminal as well as the operation of DS-100.

The HYDRA standard Windows terminal can be used as master terminal to trigger the devices "DS-100" by MPDV as well as "MT3" by IBS. This results in a data acquisition system with the following structure:

Diagram
Master Terminal and DS-100 Subbus



However, it is not possible to use both device types at one master terminal.

2 Important information

A master terminal can trigger up to 16 devices of the type "DS-100" or "MT3".

2.1 DS-100 - Overview

- One DS-100 is specifically assigned to one machine.
- It provides the following interfaces for machine communication:
 - 2 counter inputs
 - 1 relay
 - 1 digital input
- DS-100 provides the following functions:
 - Import of machine signals (quantities/cycles, malfunction signals)
 - Input of downtime reasons
 - Operation of the production lock
- Staff as well as operations cannot be logged on to DS-100 terminals. These functions need to be performed at the master terminal.
- Due to its hardware properties, the DS-100 does not provide for an exact determination of actual cycles. This can only be achieved by using CT-MSS (stored program control, "machine interface"). Please contact MPDV Project Management for further information.

2.2 MT3 - Overview

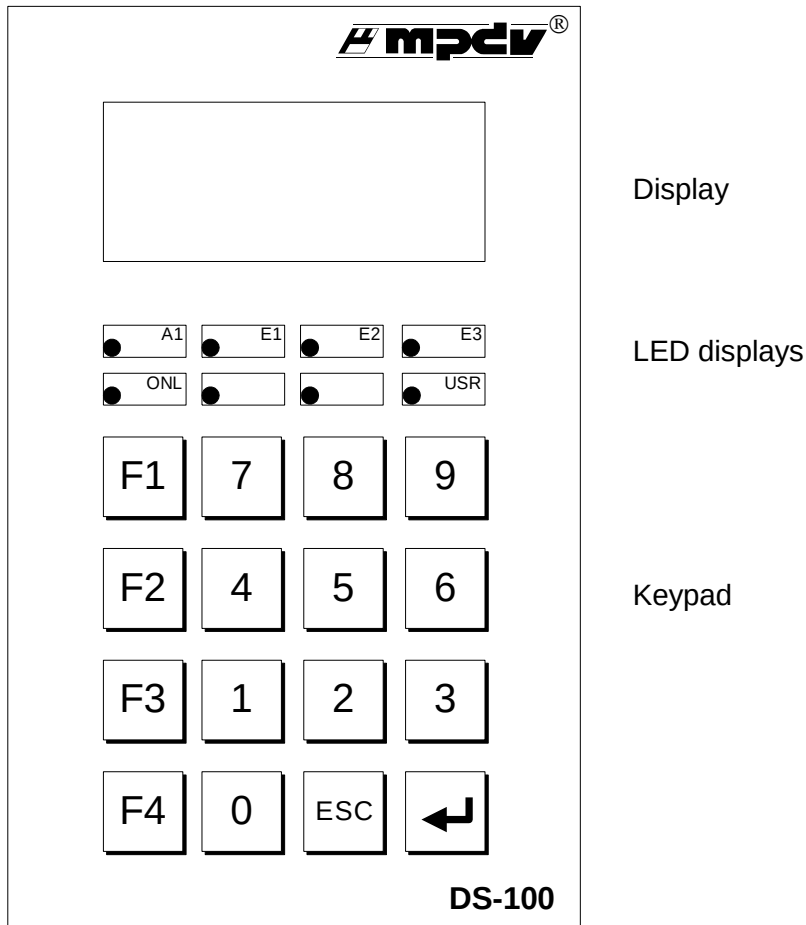
- An MT3 is exactly assigned one machine. Quantities are recorded using the machine interface (MSS).
- **Please note:** The "production" status must have a number > 20 for MT3 devices.

3 Operation

3.1 Master terminal

The functions of the HYDRA standard Windows terminal are complemented by the "service display" (see below).

3.2 Layout of DS-100



3.2.1 Functions of LED displays

A1	The display flashes if the internal relay "machine lock" is active. The configuration is performed at the HYDRA client (<i>ADE/MDE menu: Master data → Machine configuration → Machines/workplaces; tab MDE configuration → Inputs/outputs; Option</i>
E1	The display flashes if "counter 1" counts a pulse. "Master data→Machine/WP→MDE configuration--> yield counter=1" or

	"Master data→Machine/WP→MDE configuration --> scrap counter=1"
E2	The display flashes if "counter 2" counts a pulse. Configuration: like 'E1', but set counter to "2".
E3	The display flashes if the digital input of the terminal is set. This input has to be defined with "number 1" in the configuration of the HYDRA client.
LED1 / ONL	The display flashes shortly if the master terminal operates this device while polling.
LED2	The display flashes if the production lock is enabled for the machine.
LED3	Not assigned
LED4 / USR	The display flashes if the device waits for user input. e.g.: a machine status needs to be assigned

3.2.2 Keypad

F1	Enables input of a new machine status. The input fields are emptied.
F2	Not assigned
F3	Not assigned
F4	Switches the production lock.
0..9	Downtime: Input of the status number for status assignment Production: No input possible
↵	Confirmation of the input → Status number is transferred to the master terminal
ESC	Function of a reset button (the figure entered last is deleted)

3.2.3 Display

The display includes four lines with 16 characters each.

Description:

Status	Line1	Line2	Line3	Line4
Production / OP logged on	Order no. + OP no.	Yield	Scrap	"Production"
Production without OP	(free)	Yield	Scrap	"Production:___"
Downtime / OP logged on	Order no. + OP no.	"Downtime"	Current status	"Status:___"
Downtime / no OP logged on	(free)	"Downtime"	Current status	"Status:___"
Downtime / production lock active	Order no. + OP no. / (free)	"Downtime"	Current status+ "PSP"	"Status:___"
Downtime / wrong status input	Order no. + OP no. / (free)	"Downtime"	Not available	"Status:___"
Shift break	(free)	"Downtime"	"No shift"	(free)

Examples:

Production / no order logged on

```
Yield: 1234
Scrap: 33
Production
```

Production / Order logged on

```
123456789012345
Yield: 1234
Scrap: 33
Production
```

Downtime / no status assigned

```
Downtime
Not assigned
Status:___
```

Downtime / no status assigned / production
lock active

```
Downtime
Not assigned PSP
Status:___
```

Downtime / status assigned

```
Downtime
Setup      PSP
Status:___
```

Please note: Display order/operation:

In case more than one operation is still logged on after an interruption/logoff, the "next" operation displayed on the DS-100 is incidental.

3.2.4 Configuration of the DS-100 device address

- Pull out and plug in again the 9-pole bus connector.

- Click the F1 button within two seconds --> the current device address will be displayed in the **hex format** top left.
- By clicking the buttons F1 and F2, the displayed address will be increased or reduced.
- The displayed address is taken over by clicking "↵".
- The bus connector must unplugged briefly to save the new device address.

Please note for the device address:

- The terminal is configured in hexadecimal notation
- The HYDRA client is configured in decimal notation

To explain this, the table below compares figures in decimal and hexadecimal notation:

Dec. :	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Hex. :	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
Dec. :	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
Hex. :	10	11	12	13	14	15	16	17	18	19	1A	1B	1C	1D	1E	1F
Dec. :	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47
Hex. :	20	21	22	23	24	25	26	27	28	29	2A	2B	2C	2D	2E	2F
Dec. :	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63
Hex. :	30	31	32	33	34	35	36	37	38	39	3A	3B	3C	3D	3E	3F
e.g. Dec. 15 ==> Hex. 0F; Dec. 16 ==> Hex. 10.																

4 Configuration

4.1 Configuration on the HYDRA client

4.1.1 Terminal Configuration



ADE/MDE: Master data > Terminal configuration > Terminals

- Tab *General*
 - The terminals "CT760" as well as "CT830" can be used as master terminals. Consequently, the terminal type has to be set to "760" or "830".
 - The operation mode of the terminal must be set to "MDE terminal".
- Tab *MDE*
 - The *Master terminal* option must be set.

4.1.2 MDE configuration of the machine



ADE/MDE: Master data > Machine/workplace configuration > Machines/workplaces

- Tab *MDE configuration > Inputs/outputs*
 - Provided that the option "master terminal" was set in the terminal configuration, "external connection" shows the option DS-100 or MT3 as well as the device address. The device address is entered as decimal number.
 - **DS-100 only:** The relay for the machine lock is set at DS-100 if the value "1" is set in the field "machine lock". As DS-100 has only one channel, 1 must always be entered there. The relay is then set for all statuses assigned to "machine lock" as well as for the status "not assigned".

Please note: It is not possible to operate DS-100 and MT3 devices at one terminal.

4.2 Entries in the configuration file of the master terminal

The following entries are relevant for the master terminal function, in addition to the INI file settings of the Windows shop floor terminal described in the documentation dealing with the standard Windows terminal:

Entry	Comment
Section [system]	
UpdateTime=1	MDE is requested every 5 seconds by default. As polling of the assigned devices is triggered by this request, the request cycle should be reduced to one second by setting this value for the operation mode to "master terminal". (by default: 5)
Section [comports]	
Com1=MASTER	The serial interfaces used for triggering the devices, must be entered as MASTER.

5 Service display at the master terminal

The service screen can be opened by entering the shortcut "Ctrl+Alt+L" in the basic screen of the master terminal:

ID	DS/MT	Maschine	V	NET	Reset	Status(h)	E1-Z1	E2-Z2	E3	pol	FKT	Eingabe	
00	DS	6001	/	J	08.04.99 13:23	4048	65336	65535	0	1	D	3	
01	DS	6002	/	J		4044	65535	65535	0	1			
02	DS	6003	/	J		4040	65535	65535	0	1			
03	DS	6004	/	J		4042	65535	65535	0	1			
04	DS	6005	/	J		4040	65535	65535	0	1			
05	DS	6006	/	J		4042	65535	65535	0	1			
06	DS	6007	/	J		4042	65535	65535	0	1			
07	DS	6008	/	J		4042	65535	65535	0	1			
08	DS	6009	/	J		4040	65535	65535	0	1			
09	DS	6010	/	J		4042	65535	65535	0	1			
10	DS	6011	/	J		4040	65535	65535	0	1			
11	DS	6012	/	J		4040	65535	65535	0	1			
12	DS	6013	\	J		4044	65535	65535	0	1		45	
13	DS	6014	/	J		4042	65535	65535	0	1			
14	DS	6015	/	J		4042	65535	65535	0	1	D-		
15	DS	6016	/	J	08.04.99 13:05	4042	65535	65535	0	1			
			/										

Communication with the connected devices can be tracked in this view. Not all fields are filled with MT3 devices.

Field	Description
ID	Device address
DS/MT	Device type (DS-DS-100 / MT-MT3)
Machine	Machine number of the assigned machine
V	Monitoring of communication while polling: The device currently being operated shows a backslash character "\".
NET	Connection status (J: device responds / T: timeout) (not with MT3)
Reset	Start date and time are entered here if a device has been rebooted since the start of the master terminal. (Not with MT3)
Status(h)	Internal DS-100 device status (not with MT3)
E1-Z1	Meter reading of counter 1 (counted down from 65535) (not with MT3)
E2-Z2	Meter reading of counter 1 (not with MT3)

Field	Description
E3	Status of the digital input (1-set) (not with MT3)
Pol	Number of devices started during the last run (the device is started several times if it responds to a status input). (Not with MT3)
FKT	Shows the pressed function keys
Input	Shows the status value entered in the bottom line (not with MT3)